

Explaining in easy terms

Main message

SOX10 was the strongest melanoma-selective dependency: melanoma cells were strongly affected by SOX10 knockout, while most other cancer types were not. The result was broad across melanoma models, statistically robust and linked to SOX10 expression.

1. Main finding: SOX10 is a robust melanoma-selective dependency

Metric	Result	Simple interpretation
Candidate rank	1	SOX10 ranked first after applying dependency, selectivity, expression and common-essential filters.
Mean Chronos effect in melanoma	-1.369	Melanoma cells were strongly impaired when SOX10 was knocked out. More negative Chronos scores indicate stronger dependency.
Mean effect in other cancers	-0.185	The knockout effect was much smaller outside melanoma, so SOX10 was not simply essential in every cancer type.
Melanoma selectivity	-1.185	Calculated as melanoma mean minus other-cancer mean. The strongly negative value shows preferential dependency in melanoma.
Melanoma models dependent	84.0%	Most melanoma lines passed the dependency threshold, so the result was broad rather than driven by only a few models.
Other-cancer models dependent	2.8%	Very few non-melanoma lines were dependent, strengthening the melanoma-lineage specificity argument.
FDR	1.16×10^{-20}	The melanoma-versus-other-cancers difference remained extremely significant after correcting for thousands of gene tests.
Hedges' g	-4.49	The separation between melanoma and other cancers was exceptionally large, not only statistically significant.
Mean dependency probability	0.851	Across melanoma models, the average estimated probability that SOX10 was a true dependency was about 85%.
Melanoma expression prevalence	88.9%	SOX10 was expressed in most melanoma models, supporting its biological relevance as a possible target or biomarker.

Selectivity calculation: $-1.369 - (-0.185) \approx -1.185$

Important distinction: 84.0% is the fraction of melanoma models passing the dependency threshold; 0.851 is the average model-level dependency probability. They support the same conclusion but measure different things.

Expression as a potential Discovery biomarker

Melanoma models with higher SOX10 expression generally showed stronger SOX10 dependency. However, the relationship was not perfect, so expression may help identify likely responders but cannot predict dependency for every tumour.

2. What the expression–dependency analysis showed

- **SOX10 was the strongest melanoma-selective dependency.** This established the main target-discovery result.
- **Most melanoma models depended on SOX10, but not all.** The dependency is broad, yet melanoma remains biologically heterogeneous.
- **SOX10-high models generally had stronger dependency.** Higher expression was associated with more negative Chronos scores.
- **Expression could help identify a responsive subgroup.** SOX10 expression may act as a practical candidate biomarker for the SOX10-dependent state.
- **Expression is not a universal predictor.** Some high-expressing models were less dependent, so expression alone is insufficient.
- **Heterogeneity motivated the proxy-target search.** Different SOX10-dependent melanomas may rely on additional, more druggable pathways.

1	2	3	4
Strong SOX10 dependency	Expression-associated subgroup	Not all models respond equally	Search for druggable proxies

3. Summary xplanation

SOX10 ranked as the strongest melanoma-selective dependency. Its knockout strongly impaired melanoma cells but had much smaller effects in other cancer types, and 84% of melanoma models were classified as dependent. SOX10 expression was also associated with stronger dependency, suggesting that expression may help identify tumours with an active SOX10-dependent lineage programme. Because this relationship was not perfect, SOX10 expression was treated as a candidate biomarker rather than a guaranteed predictor. This heterogeneity motivated the search for more druggable co-dependencies that could provide therapeutic entry points into the SOX10-dependent melanoma state.

One-sentence takeaway: SOX10 is a strong and selective melanoma dependency, while its expression may help identify, but cannot perfectly predict, the melanoma models most reliant on that state.